

B.M.S. College of Engineering, Bengaluru-19

(Autonomous Institute, Affiliated to VTU)



Department of Electronics and Instrumentation Engineering

Mini Project –2

by

Srivatsa P (1BM21EI041)

Prajwal T (1BM21EI025)

Aditya Kumar Sinha (1BM21EI006)

Himanshu Gupta (1BM21EI012)

MENTOR:- Dr. Preethi K Mane
(Associate Professor, Dept of EIE)

AI and IoT based Smart suitcase

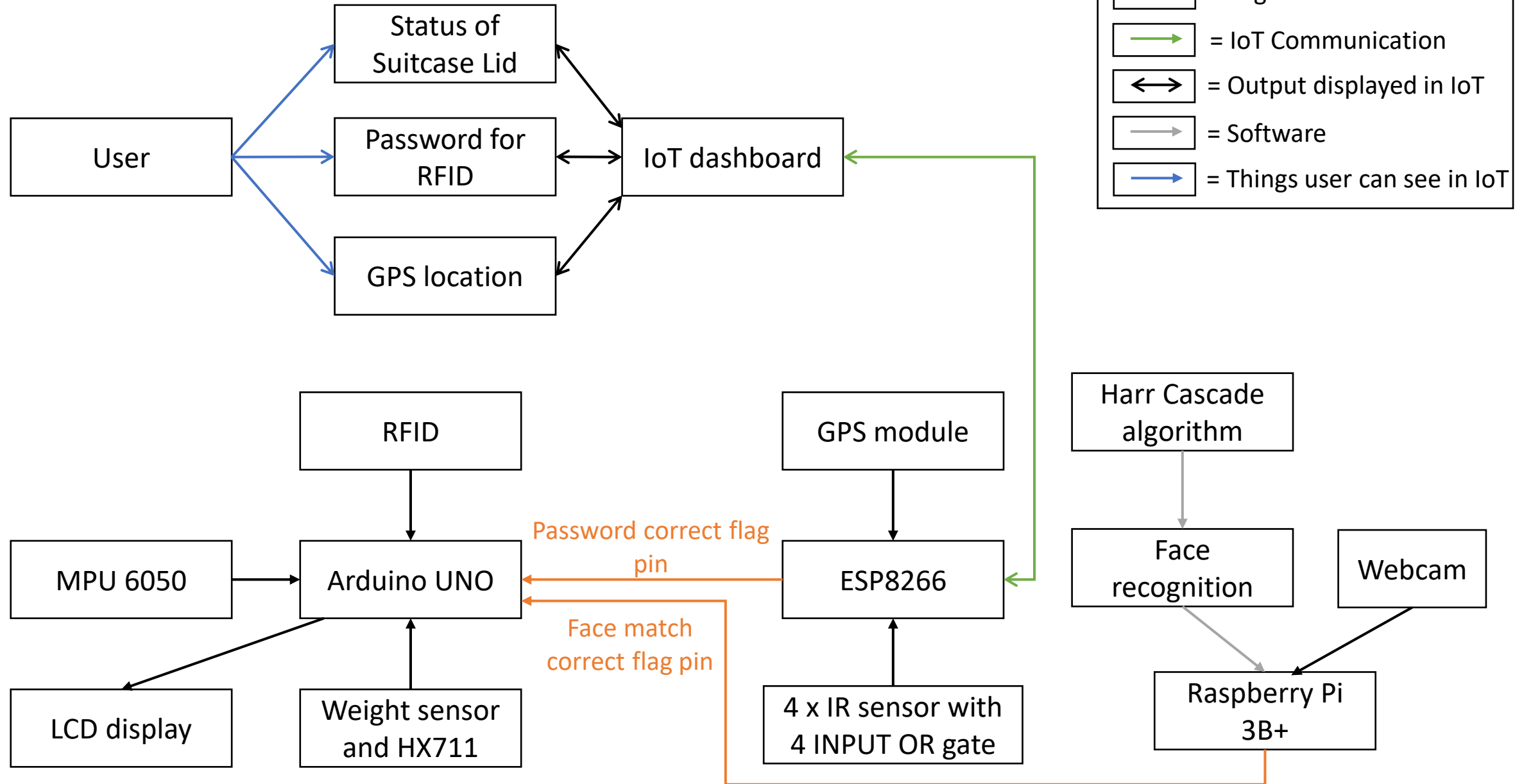
Problem Definition

Since, nowadays every items we use in daily life is getting smart with the help of modern technologies. The 'suitcase' is one of the most used items where people uses everyday. However, a traditional suitcase is used to carry our belongings, it lacks the use of modern technologies, there is a need to transform the conventional suitcase into a smart suitcase which could enhance the user experience by integrating smart devices. Current suitcases lack security concerns, weight management and several other features. By using the latest technologies, we can address the security concerns, weight management, prevention of lost luggage and provide real time monitoring of the suitcase including the GPS by also sticking to the airline regulations which mainly concerns about the battery used.

Objectives

- **Object Recognition**: To Develop an AI-based system for the camera to recognize and classify objects placed in the suitcase.
- **Weight Sensing**: Integrate a weight sensor to accurately measure the weight of items within the suitcase.
- **Security Features**: Implement security measures to detect and alert the user in case of unauthorized access or tampering.
- **GPS Tracking**: Incorporate a GPS module to enable real-time tracking and location updates of the suitcase during travel.
- **Real-time Feedback**: Provide real-time feedback to the user regarding the weight distribution and any excess baggage concerns.
- **Many access points**: To provide many access points to the suitcase if any of the access point fails and also to support the manual unlocking system with a key.

BLOCK DIAGRAM



Block diagram Explanation

- This project involved the use of three development boards: Arduino UNO, NodeMCU (ESP8266), and Raspberry Pi 3B+. The load cell, LCD, RFID module, and MPU6050 were interfaced with the Arduino UNO. Four IR sensors were connected such that their outputs were given to a 4-input OR gate, which was then interfaced with the ESP8266. The Raspberry Pi was attached to a webcam for the face recognition algorithm.
- The orange lines indicate the important digital pins used for gaining access to the suitcase. The IoT dashboard consists of three windows to view the sensor values. The first window indicates whether the suitcase lid is open or closed based on the IR sensor values. The second window displays the GPS module readings, allowing the user to track the location of the suitcase. The third window is the password section, where the user can change the password that is combined with the correct RFID authentication.

- The orange line from the Raspberry Pi indicates a digital value that is set HIGH when the face recognition system correctly recognizes a face, and is set LOW after some time. It also remains LOW when no face or an incorrect face is detected. The orange line from the ESP8266 indicates whether the entered password matches the reference password correctly.
- The LCD display shows the weight readings from the load cell, while the MPU6050 readings are used to determine the orientation of the suitcase. The MPU6050 is also used to lock the initial load cell values to prevent shifting of readings when the suitcase is in an upright position or while being moved around.
- The RFID module serves as one of the key access points, allowing the user to unlock the suitcase using the correct RFID tag. This authentication is combined with the password feature available on the IoT dashboard.

Hardware Components

Arduino UNO – Acts as the main microcontroller responsible for interfacing and coordinating sensors, access control modules, and display units in the smart suitcase system.

ESP8266 (NodeMCU) – Provides IoT connectivity, handles dashboard communication, password authentication, and transmits suitcase status data wirelessly.

Raspberry Pi 3B+ – Executes the face recognition algorithm using a connected webcam for biometric suitcase access control.

IR Sensor – Detects whether the suitcase lid is open or closed by sensing the proximity of the lid.

4-Input OR Gate (CD4072BEX H 9241 IC) – Combines the outputs of all four IR sensors to improve reliability and prevent false lid-status detection.

10KG Load Cell – Measures the weight of the luggage placed inside the suitcase.

HX711 – Amplifies and converts the low-level analog signals from the load cell into digital values for the Arduino UNO.

MPU6050 – Detects the orientation and motion of the suitcase to stabilize and lock load cell readings during movement.

Neo 6M GPS Module – Provides real-time location tracking of the suitcase for monitoring through the IoT dashboard.

Webcam – Captures facial images for the Raspberry Pi–based face recognition system.

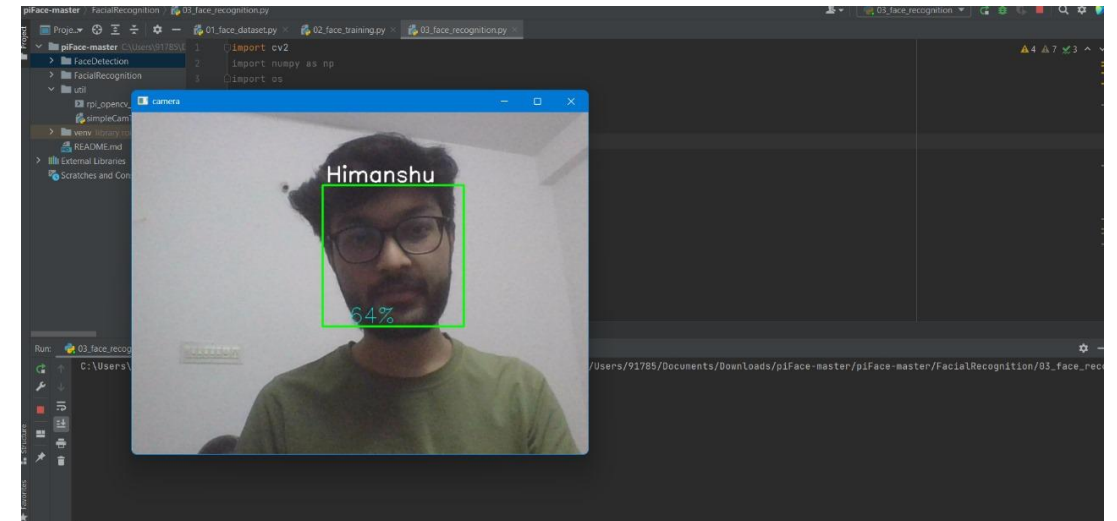
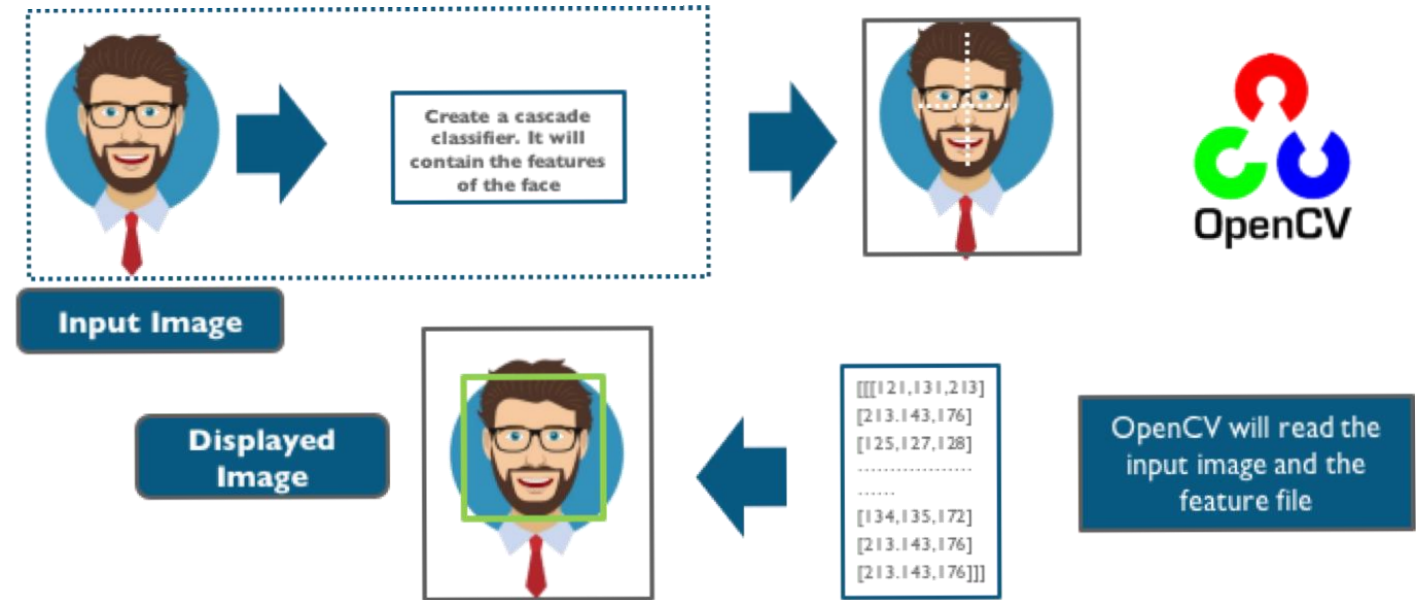
LCD Display – Displays real-time information such as suitcase weight and sensor-related status values.

Design Methodology for software

- 1) **Raspberry pi imager:** OS for the raspberry pi to install.
- 2) **Arduino IoT cloud:** To design a dashboard where it shows all the sensor's status, shows GPS and to change the password of RFID.
- 3) **OpenCV:** To write a python code for implementing AI based face recognition.

Haar Cascade Algorithm

- Haar cascade is an algorithm that can detect objects in images, irrespective of their scale in image and location
- This algorithm is not so complex and can run in real-time.
- We can train a haar-cascade detector to detect various objects like persons, cars, bikes, buildings, fruits, etc.



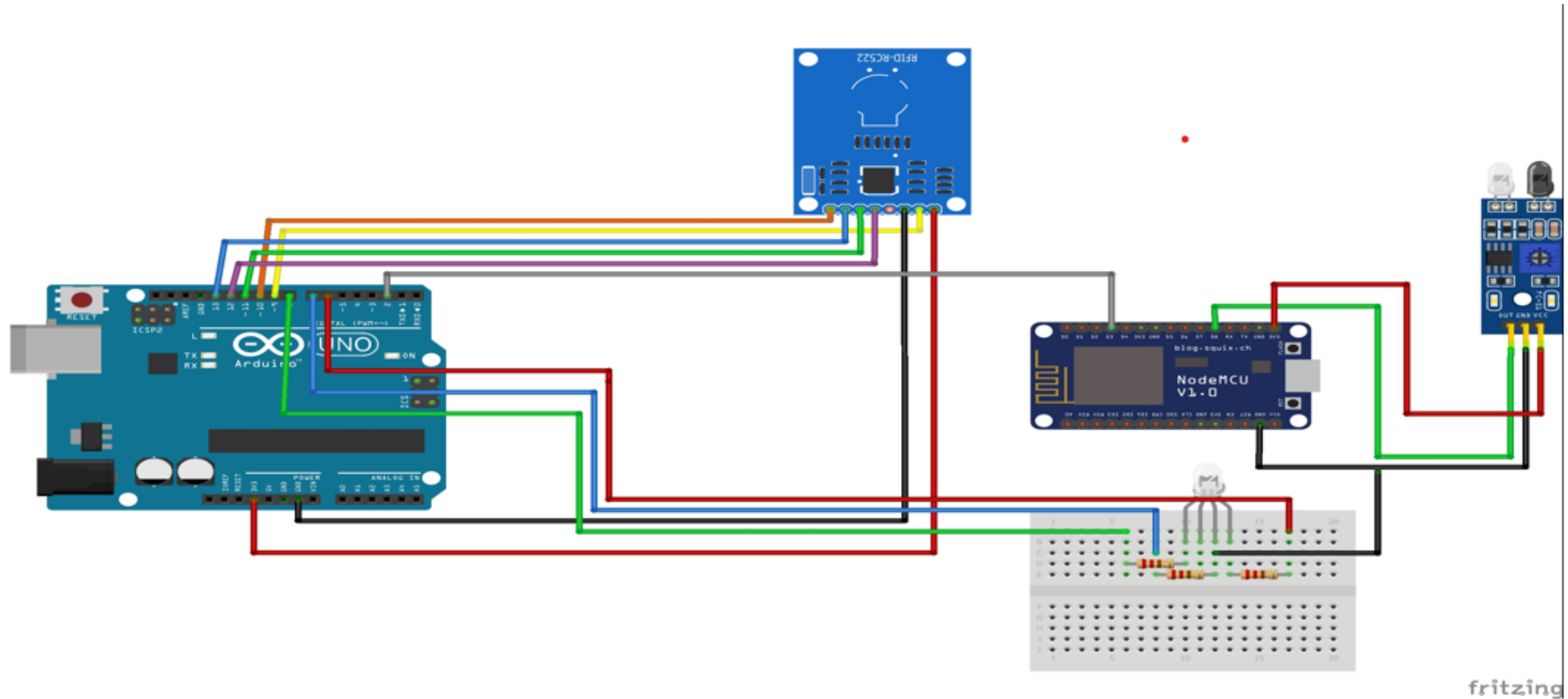
WORKING OF THE PROJECT

- The suitcase has three access points. One access point is the normal manual key unlocking mechanism, another is face recognition, which unlocks only when the correct face is recognized, and the third is RFID-based unlocking. The suitcase opens only with the correct RFID tag, which is also paired with the IoT dashboard's password mechanism.
- This acts as a safety feature in situations where the suitcase's security is threatened and the correct RFID tag is stolen. Basically, there are two strings that are initially matched with each other. When the suitcase security is compromised, the user can change the password through the IoT dashboard. Once the password is changed, the password string entered by the user and the reference password become mismatched. As a result, the ESP8266 sends a LOW signal (orange line in the block diagram) to the Arduino UNO. Due to this, even with the correct RFID tag, the suitcase will not unlock.

- When the suitcase is safe and back with the user, the user can press the button on the dashboard indicating that the suitcase is secure again. Once the button is pressed, the reference string is updated to match the password entered by the user. Since both strings now match again, the ESP8266 sends a HIGH signal to the Arduino UNO, allowing the RFID-based unlocking mechanism to function normally again.
- The load cell is used to indicate how much load has been placed inside the suitcase. This feature is included to prevent unnecessary luggage weight that may later need to be discarded at airports due to weight limitations. The MPU6050 is used to determine the orientation of the suitcase. As the suitcase moves or is placed in an upright position, severe inaccuracies can appear in the load cell readings displayed on the LCD. Therefore, when the MPU6050 detects that the suitcase is moving or is in an upright position, it locks the weight value displayed on the LCD to the last stable value recorded when the suitcase was lying flat and stationary. Once the suitcase returns to a resting and flat position, the sensor unlocks the display values and allows the load cell readings to update again.

- The IR sensors are used to detect whether the suitcase lid is open or closed. To prevent false reporting due to sensor failure or obstruction of a sensor by the suitcase contents, a total of four IR sensors are used. The outputs of all four IR sensors are connected to an OR gate. An OR gate was chosen because the IR sensor outputs are inverted in logic. When an IR sensor detects an object, it outputs a LOW signal (0), and when no object is detected, it outputs a HIGH signal (1). Therefore, the OR gate output becomes LOW only when all four IR sensor outputs are LOW, indicating that the suitcase lid is properly closed. Even if one sensor outputs HIGH, the system still reports that the suitcase is not completely closed. This status change is reflected in real time on the IoT dashboard.
- The GPS module provides the real-time location of the suitcase, and the IoT dashboard displays a map that allows the user to monitor the suitcase's movements.

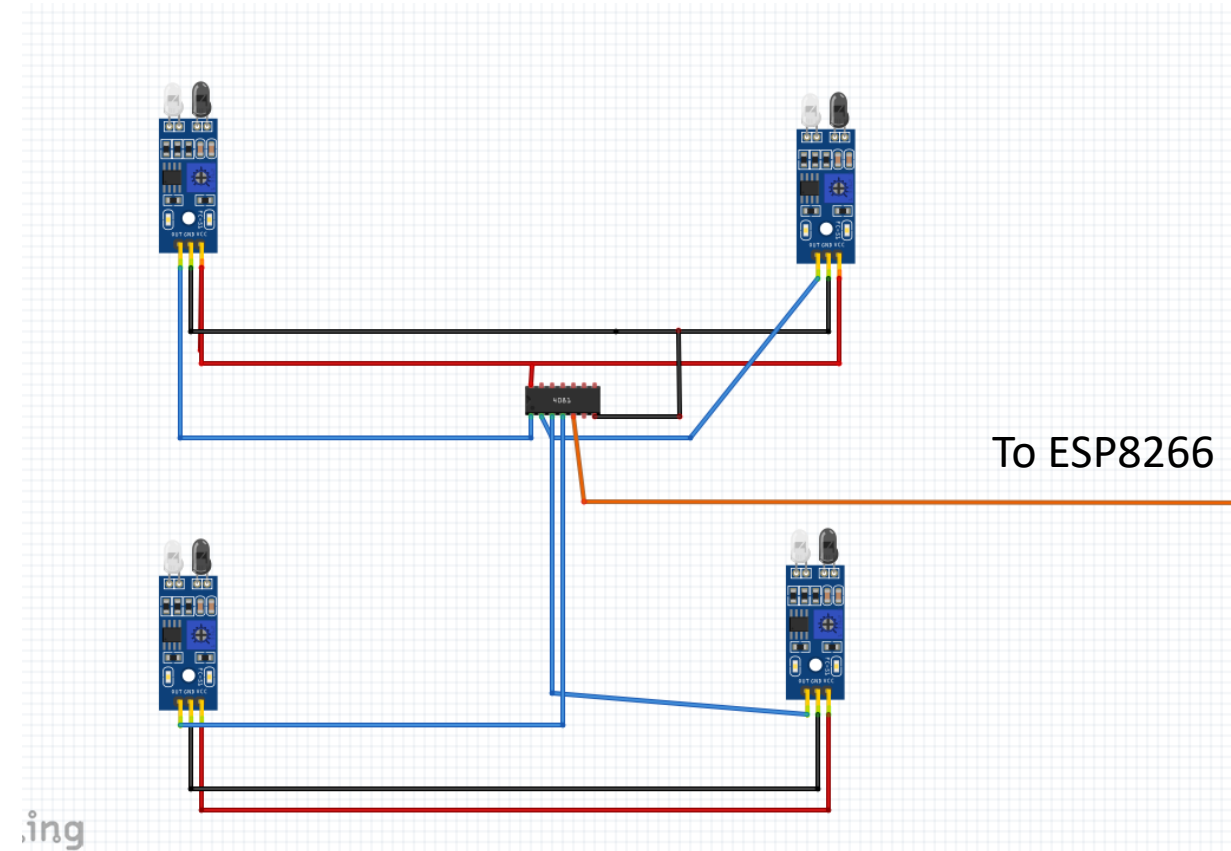
CIRCUIT DIAGRAM OF RFID WITH UNO AND NODEMCU



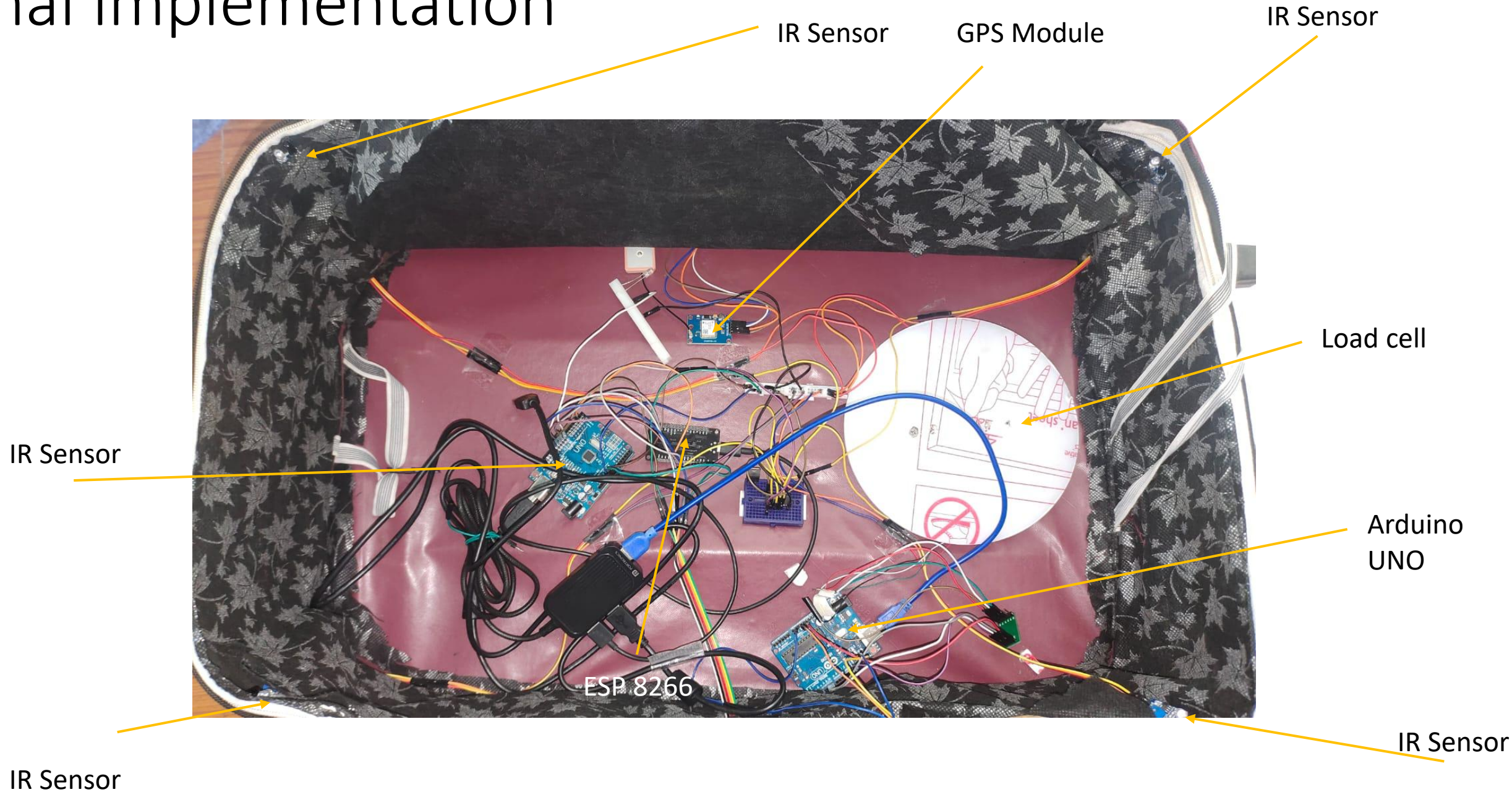
PRELIMINARY WORK MODELLING

IR sensor: -

- Must provide signal level 'HIGH' from all the IR sensors to be sure about the suitcase's lid status.
- External design of 4 INPUT OR gate is necessary.
- This configuration reduces false positive and also the above configuration is necessary to reduce the usage of pins on esp8266.
- The IC shown in the wiring diagram is for reference purposes only and was not used in the final implementation. A different IC was used in the actual hardware design.



Final implementation



THANK YOU